

Topics:

- Vectors equation of line in 3D
- Distance from a point to a line in 3D

Food for thought

- What is x -intercept?
- What is y -intercept?

Line in 2D (plane) and in 3D (Space)

A Line in plane (2D):

A **line** in the **xy -plane** is determined when **y -coordinate of a point** (that is **y -intercept**) on the line and the **direction of the line** (its slope or angle of inclination) are **given**. The **equation of the line** can then be written using the **point-slope form**.

Recall: Point-slope form:

For any point (x_0, y_0) on the line having slope m is given by the equation

$$y - y_0 = m(x - x_0)$$

If we fix the point, to be any point on the y -axis (y -intercept), then $x_0 = 0$ and $y_0 = c$, $\forall c \in \mathbb{R}$ and above equation becomes

$$y - c = mx$$

or

$$y = mx + c$$

A Line in a Space (3D):

A line L in **three-dimensional space** is determined when we know a point $P_0(x_0, y_0, z_0)$ on L and the direction of line L . In three dimensions, the direction of a line is conveniently described by a vector, so we let $\vec{v}$ be a vector parallel to the line L .

Vector Equation for a Line in space:

Let $P(x, y, z)$ be an arbitrary point on L and let $\vec{r}_0$ and $\vec{r}$ be the position vectors of P_0 and P (that is, they have representations $\overrightarrow{OP_0}$ and $\overrightarrow{OP}$). If $\vec{u}$ is the vector with representation $\overrightarrow{P_0P}$, as in **Figure 1**, then the Triangle Law for vector addition gives

$$\vec{r} = \vec{r}_0 + \vec{u}$$

But, since $\vec{u}$ and $\vec{v}$ are parallel vectors, there is a scalar $t \in (-\infty, +\infty)$ such that $\vec{u} = t\vec{v}$. Thus

$$\vec{r} = \vec{r}_0 + t\vec{v} \quad (1)$$

which is a **vector equation** of a line L .

Each value of the parameter t gives the position vector $\vec{r}$ of a point on L . In other words, as t varies, the line is traced out by the tip of the vector $\vec{r}$. As **Figure 2** indicates, positive values of t correspond to points on L that lie on one side of P_0 .

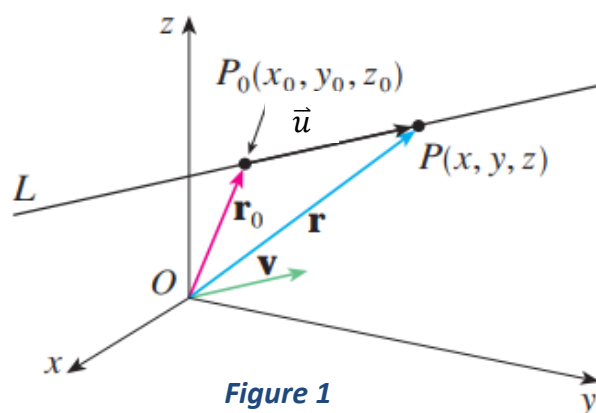


Figure 1

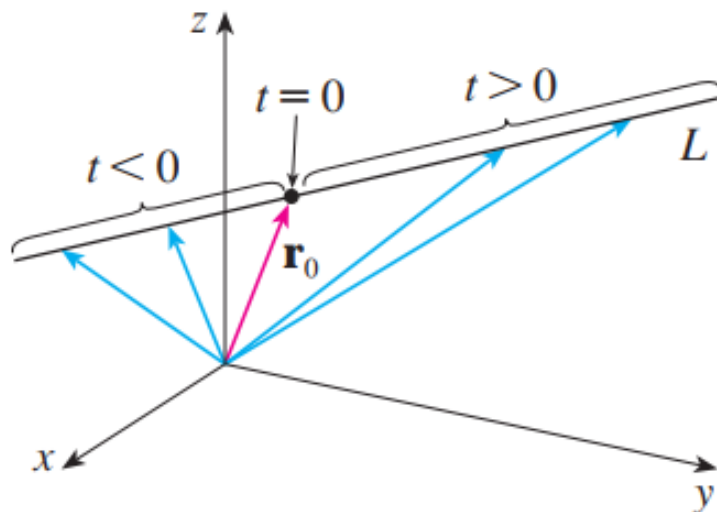


Figure 2

If the vector $\vec{v}$ that gives the direction of the line L is written in component form as $\vec{v} = \langle a, b, c \rangle$, then we have $t\vec{v} = \langle ta, tb, tc \rangle$.

We can also write $\vec{r} = \langle x, y, z \rangle$ and $\vec{r}_0 = \langle x_0, y_0, z_0 \rangle$, so the vector equation becomes

$$\langle x, y, z \rangle = \langle x_0 + ta, y_0 + tb, z_0 + tc \rangle$$

Two vectors are equal if and only if corresponding components are equal. Therefore, we have the three scalar equations:

$$x = x_0 + ta, \quad y = y_0 + tb, \quad z = z_0 + tc \quad \text{_____} (2)$$

where $t \in \mathbb{R}$.

These equations are called **parametric equations** of the line L through the point $P_0(x_0, y_0, z_0)$ and parallel to the vector $\vec{v} = \langle a, b, c \rangle$. Each value of the parameter t gives a point (x, y, z) on L .

Parametric Equation for a Line in space

Parametric equations for a line through the point $P_0(x_0, y_0, z_0)$ and parallel to the direction vector $\vec{v} = \langle a, b, c \rangle$ are

$$x = x_0 + ta$$

$$y = y_0 + tb$$

$$z = z_0 + tc$$

Note:

Two vectors $\vec{u}$ and $\vec{v}$ are parallel if $\vec{u} = t\vec{v}$, where t is scalar. We understand it with the help of an example given below.

$$\vec{u} = \langle 1, 2, 3 \rangle$$

$$\vec{v} = \langle 2, 4, 6 \rangle$$

Then

$$\vec{v} = 2 \langle 1, 2, 3 \rangle$$

$$\vec{v} = 2\vec{u}$$

This implies that $\vec{u}$ & $\vec{v}$ are parallel.

Example 1:

Find **parametric equation** & **vector equation** for the line through the point $(-2, 0, 4)$ and parallel to the vector

$$\vec{v} = 2\vec{i} + 4\vec{j} - 2\vec{k}$$

Solution:

Since the given point is

$$P = (x_0, y_0, z_0) = (-2, 0, 4)$$

and the vector is

$$\vec{v} = a\vec{i} + b\vec{j} + c\vec{k} = v_1\vec{i} + v_2\vec{j} + v_3\vec{k} = 2\vec{i} + 4\vec{j} - 2\vec{k}$$

Here we have

$$x_0 = -2, \quad y_0 = 0, \quad z_0 = 4$$

$$a = 2, \quad b = 4, \quad c = -2$$

Parametric equation of line in space:

$$x = x_0 + ta = -2 + 2t$$

$$y = y_0 + tb = 0 + 4t = 4t$$

$$z = z_0 + tc = 4 - 2t$$

That is,

$$x = -2 + 2t$$

$$y = 4t$$

$$z = 4 - 2t$$

where $-\infty < t < \infty$.

Vector Equation of Line in space:

$$\vec{r} = \vec{r}_0 + t \vec{v}$$

$$x\vec{i} + y\vec{j} + z\vec{k} = (-2\vec{i} + 0\vec{j} + 4\vec{k}) + t(2\vec{i} + 4\vec{j} - 2\vec{k}); \quad -\infty < t < \infty$$

$$x\vec{i} + y\vec{j} + z\vec{k} = (-2\vec{i} + 0\vec{j} + 4\vec{k}) + (2t\vec{i} + 4t\vec{j} - 2t\vec{k})$$

$$x\vec{i} + y\vec{j} + z\vec{k} = (-2 + 2t)\vec{i} + (0 + 4t)\vec{j} + (4 - 2t)\vec{k}$$

$$x\vec{i} + y\vec{j} + z\vec{k} = (-2 + 2t)\vec{i} + (4t)\vec{j} + (4 - 2t)\vec{k}$$

Example 2:

Find the **parametric equation** of the line passing through origin and parallel to the vector

$$\vec{v} = 2\hat{j} + \hat{k}$$

Solution:

Given that

$$\text{Origin} = P = (0,0,0) = P(x_0, y_0, z_0)$$

$$\vec{v} = 0\hat{i} + 2\hat{j} + \hat{k} = a\hat{i} + b\hat{j} + c\hat{k}$$

Then the parametric equations of line are

$$x = x_0 + at = 0 + 0t = 0$$

$$y = y_0 + bt = 0 + 2t = 2t$$

$$z = z_0 + ct = 0 + t = t$$

Then the required parametric equations are

$$\mathbf{x = 0}$$

$$\mathbf{y = 2t}$$

$$\mathbf{z = t}$$

Example 3:

Find the **parametric equation** and **vector equation** of the line through $P(-3, 2, -3)$ and $Q(1, -1, 4)$.

Solution:

$$P = (x_0, y_0, z_0) = (-3, 2, -3)$$

$$Q = (x_1, y_1, z_1) = (1, -1, 4)$$

The vector parallel to line is

$$\vec{v} = \overrightarrow{PQ} = \langle x_1 - x_0, \quad y_1 - y_0, \quad z_1 - z_0 \rangle$$

$$\vec{v} = \overrightarrow{PQ} = \langle 1 - (-3), \quad -1 - 2, \quad 4 - (-3) \rangle$$

$$\vec{v} = \overrightarrow{PQ} = \langle 4, -3, 7 \rangle = \langle a, b, c \rangle$$

We can write the vector in standard form as

$$\vec{v} = a\vec{i} + b\vec{j} + c\vec{k} = 4\vec{i} - 3\vec{j} + 7\vec{k}$$

Parametric equation of a line in space:

$$x = x_0 + t v_1 = -3 + 4t$$

$$y = y_0 + t v_2 = 2 - 3t$$

$$z = z_0 + t v_3 = -3 + 7t$$

where $-\infty < t < \infty$.

Vector equation for a line in space:

$$xi + yj + zk = (-3i + 2j - 3k) + t (4i - 3j + 7k), \quad -\infty < t < \infty$$

$$xi + yj + zk = (-3i + 2j - 3k) + (4t i - 3t j + 7t k)$$

$$xi + yj + zk = (-3 + 4t) i + (2 - 3t)j + (-3 + 7t)k$$

Do yourself Exercise 12.5 (Textbook: Thomas Calculus 11th Edition):

Q # 1 to 20.

Example 4:

Find the **parametric equation** of the line passing through the point $(3, -2, 1)$ and parallel to the line having parametric equations

$$x = 1 + 2t$$

$$y = 2 - t$$

$$z = 3t$$

Solution:

Since we have given the parametric equations as

$$x = 1 + 2t = x_0 + at$$

$$y = 2 - t = y_0 + bt$$

$$z = 0 + 3t = z_0 + ct$$

This implies that

$$a = 2, \quad b = -1, \quad c = 3$$

which are the components of a vector **v** , given as

$$\mathbf{v} = \langle \mathbf{a}, \mathbf{b}, \mathbf{c} \rangle = \langle \mathbf{2}, \mathbf{-1}, \mathbf{3} \rangle$$

The given point is

$$P = (3, -2, 1) = P(x_0, y_0, z_0)$$

$$\mathbf{v} = \langle \mathbf{a}, \mathbf{b}, \mathbf{c} \rangle = \langle \mathbf{2}, \mathbf{-1}, \mathbf{3} \rangle$$

That is,

$$x_0 = 3, \quad y_0 = -2, \quad z_0 = 1$$

$$a = 2, \quad b = -1, \quad c = 3$$

Then the **parametric equation** of the line passing through the point $P(3, -2, 1)$ and parallel to the line with the given parametric equations are given below:

$$x = x_0 + at$$

$$y = y_0 + bt$$

$$z = z_0 + ct$$

Putting the values, we have

$$x = 3 + 2t$$

$$y = -2 - t$$

$$z = 1 + 3t$$

which are the required parametric equations of the line.

Example 5:

Find the parametric equations of the line passing through the point $(2, 3, 0)$, and perpendicular to the vectors $\vec{u} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{v} = 3\hat{i} + 4\hat{j} + 5\hat{k}$.

Solution:

Since the line is perpendicular to the vectors $\vec{u}$ and $\vec{v}$, therefore using the cross product, we have

$$\vec{u} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ 3 & 4 & 5 \end{vmatrix}$$

$$\vec{u} \times \vec{v} = \hat{i}(10 - 12) - \hat{j}(5 - 9) + \hat{k}(4 - 6)$$

$$\vec{u} \times \vec{v} = -2\hat{i} + 4\hat{j} - 2\hat{k} = a\hat{i} + b\hat{j} - c\hat{k}$$

The given point is

$$P = (2, 3, 0) = P(x_0, y_0, z_0)$$

Then the parametric equations of line are

$$x = x_0 + at = 2 - 2t$$

$$y = y_0 + bt = 3 + 4t$$

$$z = z_0 + ct = 0 - 2t$$

Then the required parametric equations are

$$x = 2 - 2t$$

$$y = 3 + 4t$$

$$z = -2t$$

Practice questions:

(Textbook: Thomas Calculus 11th Edition) Ex. 12.5: 1-7, 10.

Parameterization of a Line segment

Example:

Parameterize the line segment joining the points $P(-3, 2, -3)$ to $Q(1, -1, 4)$.

Solution:

First of all, we will find the parametric equation of the line through the points $P(-3, 2, -3)$ and $Q(1, -1, 4)$ and then restrict the domain of parameter t to obtain the parametric equation of the line segment from P to Q .

Step 1: (Equation of line)

$$\begin{aligned}\vec{v} = \overrightarrow{PQ} &= (1 + 3)\hat{i} + (-1 - 2)\hat{j} + (4 + 3)\hat{k} \\ \vec{v} &= 4\hat{i} - 3\hat{j} + 7\hat{k} = a\hat{i} + b\hat{j} + c\hat{k}\end{aligned}$$

Let us consider the point $P(-3, 2, -3) = P(x_0, y_0, z_0)$. (NOTE: We can also consider point Q here). Then the parametric equations of line are

$$\begin{aligned}x &= x_0 + at = -3 + 4t \\ y &= y_0 + bt = 2 - 3t \\ z &= z_0 + ct = -3 + 7t\end{aligned}$$

Step 2: (Line segment)

In order to find the value of t for which an arbitrary point (x, y, z) of the line is at $P(-3, 2, -3)$, we solve the equation

$$\left. \begin{aligned} -3 &= -3 + 4t \\ 2 &= 2 - 3t \\ -3 &= -3 + 7t \end{aligned} \right\} \Rightarrow t = 0$$

Similarly, when (x, y, z) is at $Q(1, -1, 4)$, we solve

$$\left. \begin{aligned} 1 &= -3 + 4t \\ -1 &= 2 - 3t \\ 4 &= -3 + 7t \end{aligned} \right\} \Rightarrow t = 1$$

So, the parametric equation of the line segment is

$$x = -3 + 4t$$

$$y = 2 - 3t$$

$$z = -3 + 7t; \quad 0 \leq t \leq 1$$

Question 19: Find the parametric equations of the line segment joining the points $P(-2, 0, 2)$ and $Q(0, 2, 0)$.

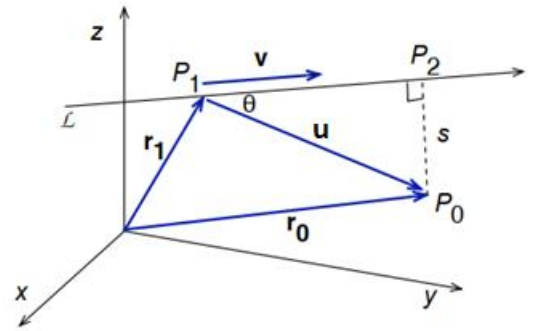
Practice Questions

(Textbook: Thomas Calculus 11th Edition) Ex. 12.5: 13-20.

The Distance from a Point to a Line in Space

Let L be a line and let $\vec{v}$ be a vector parallel to the line L . This means that $\vec{v} \parallel L$.

Let P_0 be a point in space from which we need to measure the distance to the line L , i.e., we need to measure the distance between P_0 and L .



We consider an arbitrary point P_1 on the line L assuming it might give us a distance from the point P_0 .

Considering P_0 and P_1 , we draw their **position vectors** $\vec{r}_0$ and $\vec{r}_1$ respectively from the **origin**.

We draw a vector $\vec{u}$ at an angle θ from P_1 to P_0 such that by **head to tail rule** we have,

$$\vec{r}_1 + \vec{u} = \vec{r}_0$$

This implies that

$$\vec{u} = \vec{r}_0 - \vec{r}_1.$$

Now logically the shortest distance of a point from a line is the length of the perpendicular drawn from the point to the line.

Let us consider a point P_2 on the line L , such that P_0P_2 is perpendicular to the line L . i.e., $P_0P_2 \perp L$. We say that $\overline{P_0P_2} = s$.

Distance is a scalar quantity. From the figure, we can write

$$\sin \theta = \frac{\text{Perpendicular}}{\text{Hypotenuse}} = \frac{s}{|\vec{r}_0 - \vec{r}_1|}$$

$$\sin \theta = \frac{s}{|\vec{u}|}$$

$$|\vec{u}| \sin \theta = s$$

$$s = |\vec{u}| \sin \theta$$

Here θ is the angle between $\vec{u}$ and $\vec{v}$. $P_0P_2 \perp L$ and $\vec{v} \parallel L$ implies that $P_0P_2 \perp \vec{v}$.

Now considering $\vec{v}$ and $\vec{u}$, we have

$$\vec{u} \times \vec{v} = |\vec{u}| |\vec{v}| \sin \theta$$

We can rearrange as

$$\vec{u} \times \vec{v} = |\vec{v}| |\vec{u}| \sin \theta$$

We know $s = |\vec{u}| \sin \theta$

$$\vec{u} \times \vec{v} = |\vec{v}| s$$

or

$$\vec{u} \times \vec{v} = s |\vec{v}|$$

Since distance is a scalar quantity thus, we need to take the magnitude of the cross product to balance the equation.

$$|\vec{u} \times \vec{v}| = s |\vec{v}|$$

$$\frac{|\vec{u} \times \vec{v}|}{|\vec{v}|} = s$$

This implies that the distance between the line L and a point can be computed by the formula,

$$s = \frac{|\vec{u} \times \vec{v}|}{|\vec{v}|}$$

where $\vec{u} = \vec{r_0} - \vec{r_1}$

Example:

Find the distance from the point $S(1, 1, 5)$ to the line:

$$L: \begin{cases} x = 1 + t \\ y = 3 - t \\ z = 2t \end{cases}$$

Solution:

Since the given point is $S(x_0, y_0, z_0) = P_0(1, 1, 5)$.

And the given parametric equations of line are

$$\begin{aligned} x &= x_1 + at = 1 + t \\ y &= y_1 + bt = 3 - t \\ z &= z_1 + ct = 0 + 2t \end{aligned}$$

This means that $P_1(x_1, y_1, z_1) = P_1(1, 3, 0)$ and $\vec{v} = \langle a, b, c \rangle = \langle 1, -1, 2 \rangle$.
Thus, the vector parallel to the line L is

$$\vec{v} = \hat{i} - \hat{j} + 2\hat{k}$$

The Line passes through the point $P_1(1, 3, 0)$

$$\vec{u} = \overrightarrow{P_1P_0} = \overrightarrow{OP_0} - \overrightarrow{OP_1}$$

$$\vec{u} = \overrightarrow{P_1P_0} = (\hat{i} + \hat{j} + 5\hat{k}) - (\hat{i} + 3\hat{j} + 0\hat{k})$$

$$\vec{u} = (1 - 1)\hat{i} + (1 - 3)\hat{j} + (5 - 0)\hat{k}$$

$$\vec{u} = 0\hat{i} - 2\hat{j} + 5\hat{k}.$$

Now,

$$\vec{u} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -2 & 5 \\ 1 & -1 & 2 \end{vmatrix}$$

$$\vec{u} \times \vec{v} = \hat{i} \begin{vmatrix} -2 & 5 \\ -1 & 2 \end{vmatrix} - \hat{j} \begin{vmatrix} 0 & 5 \\ 1 & 2 \end{vmatrix} + \hat{k} \begin{vmatrix} 0 & -2 \\ 1 & -1 \end{vmatrix}$$

$$\vec{u} \times \vec{v} = \hat{i}(-4 + 5) - \hat{j}(0 - 5) + \hat{k}(0 + 2)$$

$$\vec{u} \times \vec{v} = \hat{i} + 5\hat{j} + 2\hat{k}$$

Taking magnitude of $\vec{u} \times \vec{v}$, we have

$$|\vec{u} \times \vec{v}| = \sqrt{(1)^2 + (5)^2 + (2)^2}$$

$$|\vec{u} \times \vec{v}| = \sqrt{30}$$

Similarly, taking magnitude of vector $\vec{v}$, we have

$$|\vec{v}| = \sqrt{(1)^2 + (-1)^2 + (2)^2} = \sqrt{6}$$

Now, apply the formula, we have

$$d = \frac{|\vec{u} \times \vec{v}|}{|\vec{v}|}$$

Putting values, we have

$$d = \frac{\sqrt{30}}{\sqrt{6}} = \sqrt{\frac{30}{6}} = \sqrt{5}$$

$$d = \sqrt{5} = 2.24$$

Hence the distance of a point P_1 to the line L is **2.24** units.

Practice Questions:

Textbook: Thomas Calculus 11th Edition Ex. 12.5: 33-38

In Exercises 33–38, find the distance from the point to the line.

33. $(0, 0, 12); \quad x = 4t, \quad y = -2t, \quad z = 2t$

34. $(0, 0, 0); \quad x = 5 + 3t, \quad y = 5 + 4t, \quad z = -3 - 5t$

35. $(2, 1, 3); \quad x = 2 + 2t, \quad y = 1 + 6t, \quad z = 3$

36. $(2, 1, -1); \quad x = 2t, \quad y = 1 + 2t, \quad z = 2t$

37. $(3, -1, 4); \quad x = 4 - t, \quad y = 3 + 2t, \quad z = -5 + 3t$

38. $(-1, 4, 3); \quad x = 10 + 4t, \quad y = -3, \quad z = 4t$